# GAURAV BANGADE

**Product Engineer · React, TypeScript & Node.js · Frontend Platforms & Offline-First Systems**

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## SUMMARY

Software engineer with 2 years on a multi-tenant maritime emissions-compliance SaaS, working across the whole product. One of the main frontend authors of the production app (960+ merged UI pull requests across 9 releases) and the designer of the company's next platform and its design system. I also lead the work that keeps the product shippable: type safety, security, release stabilisation and documentation. Strongest in React and TypeScript; comfortable owning the API in NestJS or Django REST Framework.

## EXPERIENCE

**Software Engineer | ecoSAIL Infotech, Pune** Nov 2024 – Present

- **Production frontend:** Built core features of **Ecosailer**, used by ship crews and shore offices: **reporting wizards** that work on phones and tablets, **analytics dashboards**, user management and a trial period on a server-owned clock.
- **Voyage Estimator:** Built it end to end, UI and API: it estimates a voyage's **$CO_2$, CII rating and EU ETS cost** before it sails. Moved every regulatory constant into the database, so rule changes ship as **data, not a deploy**. 55 tests.
- **Platform architecture (in development):** Designed **CVMS**, the platform future products will be built on: one NestJS API and one React app assembled from **module manifests**, with module boundaries **enforced by lint and CI**. The channel-partner portal already runs on its patterns.
- **Offline-first sync (in development):** Built the sync foundation for vessels on slow, unreliable **satellite links** (transactional outbox, hybrid logical clock) and a **test harness** that checks every synced row under **10 simulated network profiles** and clock skew.
- **Design system:** Created **Ensign**, a token-based design system with light and dark themes. It is now the required UI layer for **2 products**.
- **Access control & security:** Rebuilt **RBAC** so nobody can grant more access than they hold and every route is **denied by default**; moved auth tokens to **httpOnly cookies**, added XSS sanitisation and audited all **4 company repos**.
- **Real-time helpdesk:** Built the shore hub's UI from scratch (**React 19**, TanStack Query) and its helpdesk: ticket workflows, **webhooks**, API keys and live **WebSocket** notifications.
- **Code quality & releases:** Led a 6-phase migration that removed **every "any" type** from the UI and surfaced real bugs, cut polling traffic **10×**, and closed **12 critical and 18 high-priority** defects to ship release 2.4.0.

## TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL
- **Frontend:** React 18/19, TanStack Query, Router & Table, Redux, Zustand, Zod, Tailwind CSS, Vite, ECharts
- **Backend:** Node.js, NestJS, Fastify, Prisma, Django REST Framework, Django Channels, Express
- **Data:** PostgreSQL, MySQL, Redis, MongoDB
- **Architecture:** Design systems, Modular monoliths, Offline-first sync, WebSockets, RBAC & auth security
- **Testing & tooling:** Vitest, Playwright, Testing Library, MSW, ESLint, Docker, Git
- **AI / LLM:** LangChain, RAG, FAISS, MCP
- **Domain:** Maritime emissions compliance: IMO CII, EU MRV / ETS, FuelEU

## SIDE PROJECTS

- **InquireAI** (Python, LangChain, FAISS, Streamlit, MongoDB): Ask questions of your PDFs in plain language: a RAG pipeline over a FAISS vector store returns answers grounded in the documents.
- **Artisan Studio** (React, Node.js, Express, MongoDB, Cloudinary): A full-stack marketplace for artisans — profiles, product uploads, reviews and buyer messaging.
- **JSON Toolkit** (React, TypeScript, React Flow): Format, validate and explore JSON as an interactive graph. Live on my portfolio site.

## EDUCATION

- **Master of Computer Applications (MCA)**, MES' Institute of Management & Career Courses, Savitribai Phule Pune University SGPA 8.0 | 2024
- **Bachelor of Computer Applications (BCA)**, Kamla Nehru Mahavidyalaya, Nagpur 84% | 2022

## LANGUAGES

English, Hindi, Marathi